
E-Commerce Customer Churn Prevention: A Two-Stage ML and Agentic AI Approach

Raghusrinivasan Venkatesan
rivenkatesan@ucsd.edu

Jason
kupai@ucsd.edu

Matteo Persiani
mpersiani@ucsd.edu

Ayoob Al-Delaimy
alaldelaimy@ucsd.edu

Narain Mylapore Sudhakar
nmylaporesudhakar@ucsd.edu

Allen Ekoung Keng
aekeng@ucsd.edu

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Abstract

Reducing customer churn is critical for e-commerce businesses, but predicting which customers are likely to leave and taking timely action is challenging.

1 Introduction

Customer retention is more cost-effective than acquisition, yet predicting and preventing churn remains challenging. Traditional approaches rely on manual analysis and generic campaigns. We want to present an integrated system that: (1) predicts churn using ensemble ML methods and (2) segments customers for targeted interventions

Contributions:

- End-to-end churn prevention pipeline combining supervised and unsupervised learning
- Multi-agent architecture for automated campaign generation with context engineering

2 Libraries

Data Processing: Pandas, NumPy for data manipulation; Matplotlib, Seaborn for visualization.

ML: Scikit-learn (Random Forest, K-means), XGBoost; MLflow for experiment tracking.

2.1 Timeline (5 Weeks)

Weeks	Phase	Leaders
1	Data and EDA, feature engineering, preprocessing	Matteo, Allen
2-3	Churn prediction models	Narainm, Raghu, Jason
3-4	Customer segmentation and analysis	Ayoob, Allen, Jason
4-5	Optimization and reporting	Matteo, Raghu
5	Finalization, documentation, report	Allen

Table 1: Project timeline

3 Expected Evaluation Metrics

ML Models: Precision, Recall, F1-score (target > 0.70)

F1-score: The harmonic mean of precision and recall, summarizing the trade-off between catching true churners (recall) and avoiding false alarms (precision):

$$F1 = \frac{2 (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}}.$$

System: Churn reduction rate, campaign conversion rate, false positive rate (avoiding over-contacting).

4 Expected Outcomes

1. Actionable customer segments with distinct profiles
2. Automated campaign generation system reducing manual effort by 50-70%
3. Comprehensive analysis identifying key churn drivers
4. Complete Python codebase: Jupyter notebooks + modular scripts

5 Datasets & Deliverables

Data: Kaggle/UCI e-commerce datasets (i.e. <https://www.kaggle.com/datasets/nabihazahid/e-commerce-customer-insights-and-churn-dataset>) or synthetic data via Faker (10K+ customers, 6+ months history).

Code: Jupyter notebooks (for preprocessing, prediction, segmentation, agents, evaluation); Python modules (`data_pipeline.py`, `ml_models.py`, `agents.py`).

Outputs: `high_risk_customers.csv`, `segment_profiles.json`, `campaigns.csv`, performance metrics.

Report: Analysis with visualizations, breakdowns, and recommendations.